Coulomb’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electric force to charge magnitude and separation.

SUCCESS CRITERIA - I can...

1. Identify charge signs before predicting direction.
2. Use inverse-square scaling.
3. Calculate force with charge in coulombs and distance in meters.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “charge” and “separation” to explain your answer. Sentence starters are optional.

charge: a property responsible for electric interactions

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A wave travels at fixed speed. A second target is **3** times farther away than the first. **Compare one-way travel times**.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F = k|q_1q_2|/r^2$; $k = \mathbf{9.0\times10^9\ N\cdot m^2/C^2}$.
r is separation. Signs find attraction/repulsion; the equation gives magnitude.

Worked reasoning: Doubling the separation between two fixed charges makes Coulomb-force magnitude one quarter as large. The charge signs find attraction or repulsion separately.

Your turn: Sketch both separations for two positive charges and **label the force directions**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Label charges and separation.

Object A Object B

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two opposite charges are moved twice as far apart. **Draw their force directions** and **compare the new force** with the original.

Sentence starter: Both ... . They differ in ... because ... .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

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Continue down each column.

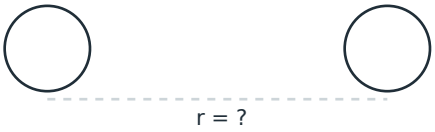
Sentence starters are optional. You may plan in any language.

04 Apply the physics

Two fixed charges attract with a force of **12 N** in a simulation. Double their center-to-center separation while keeping both charges fixed. **Label both separations** and **calculate the new force magnitude** using the inverse-square relationship.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Charge 1 Charge 2

Marzano check - circle: 1 2 3

05 Explain your reasoning

One charge has twice the magnitude of the other. Does it exert twice the force on the smaller charge? Explain.

Word help: magnitude = size of a quantity without its direction

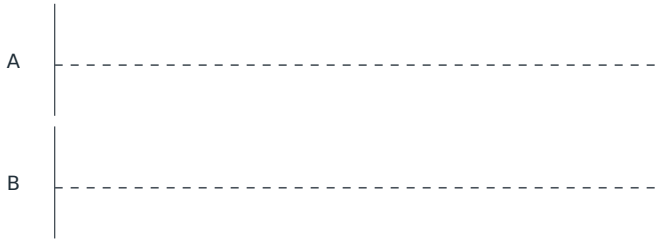
Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **600 Hz** sound travels through air. Do air particles mainly move parallel or perpendicular to propagation? **Name the wave type**.
Label a wave sketch to **support the calculation or comparison**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Time or position: choose and label

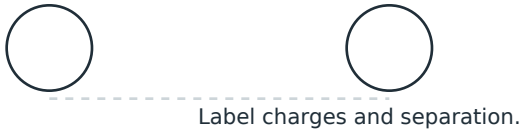
Marzano check - circle: 1 2 3

07 Exit check

With separation fixed, one charge magnitude triples. **What happens** to the force magnitude?

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice ... . This supports ... because ... .



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A force is **12 N**. Double one charge and double separation. **Find the new magnitude** and **explain each factor**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3